Exploratory Evaluation of a Single Residual-Stream Axis for Condition Labels and Binary-Report Logits in Qwen3-8B

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Dr. Ione Mercer (fictional author)

> **Disclosure:** This AI-generated manuscript was revised after peer review using the complete non-sensitive protocol, sanitized configuration, aggregate results, analysis source, layer table, and verifier source and output supplied for reconciliation. The post-run verifier is an independently implemented consistency check over frozen artifacts, not an independent scientific replication.

Abstract

This study tested whether input-relative availability is associated with a single residual-stream direction that transfers across prompt families and whether adding that direction changes a YES-minus-NO output-logit contrast more than adding an orthogonal control direction. A local `mlx-community/Qwen3-8B-4bit` checkpoint was evaluated using three training families, one validation family, and one final test family. Each family contained six paired cases: the two prompts in a pair had the same field contents, including one visible value and one masked value, but queried either the visible or masked field. This produced 36 training, 12 validation, and 12 test observations.

At every valid layer, the fitted direction was the unit-normalized difference between the positive- and negative-class means of final-token training activations. Classification used the midpoint between the two training-class mean projection scores. Layer selection maximized validation accuracy, then validation standardized mean difference, with earlier layer as the final tie-breaker. Layer 32 after the transformer block was selected with validation accuracy `0.8333333333333334` and standardized mean difference `2.378871853430071`. A 500-permutation maximum-over-layers analysis, which refitted the direction for every shuffled set of training labels, gave $p = 61/501 \approx 0.122$. This failed to reject its label-permutation null at a conventional, non-registered 0.05 threshold.

On the final packet family, the selected score classified 11 of 12 observations correctly: accuracy `0.9166666666666666`, with a conditional stratified-bootstrap 95% interval of [`0.75`, `1.0`]. The positive-minus-negative mean score gap was `76.23535315281211`, with a conditional 95% interval of [`48.30552862846039`, `99.67488431854588`]. These intervals used 2,000 within-class bootstrap resamples while holding the fitted direction, threshold, and selected layer fixed. They did not preserve the six matched case pairs and do not quantify training, selection, or prompt-family uncertainty.

For the intervention, a unit fitted direction and a unit Gaussian random direction orthogonalized against it were added separately to the final-token residual stream after block 32 and before block 33. Both used the same scale, `29.591223144531252`, equal to 5% of the median norm of the selected training residuals. Across

12 packet prompts and seven alpha values, pooled ordinary least-squares slopes per unit alpha were `1.2159963259621271` for the fitted direction and `0.04448462289477151` for the orthogonal random direction on the YES-minus-NO contrast. Their descriptive difference was `1.1715117030673556` (approximately `1.171512`). No uncertainty or distribution over control directions was computed. The fitted direction also changed the related TRUE-minus-FALSE contrast, with slope `0.45956590062096003`.

The results support a narrow claim: in this checkpoint, a unit mean-difference axis predicted the constructed condition labels in a final prompt family, and artificial addition along that axis changed designated binary token-logit contrasts more than one exactly orthogonal, equal-norm random intervention. They do not establish intrinsic low dimensionality, transfer over a population of prompt families, semantic specificity to input-relative availability, normal-use mediation, or phenomenal consciousness.

Introduction

Language models can be asked whether information required for a response is present in their current input. Some behavioral studies have interpreted success on restricted tasks of this kind as introspection-like behavior [1]. Negative findings concerning models' knowledge of linguistic properties caution against generalizing such results into broad introspective competence [2]. Neither literature establishes that the present operationalization measures a model's representation of its own epistemic access.

Four claims must remain distinct:

1. an internal score predicts an experimentally assigned report label;
2. an artificial intervention changes a designated report-token contrast;
3. natural variation along the score normally mediates the report;
4. the model possesses phenomenal consciousness.

The present experiment addresses the first claim and establishes a causal effect of a specified artificial activation addition on selected output logits. It does not establish normal-use mediation or phenomenal consciousness.

Residual-stream activation additions can steer language-model behavior [3], and representation-engineering research motivates the joint study of activation readout and control [4]. Related activation-injection work concerning introspection-like behavior supplies additional motivation while emphasizing reliability boundaries [5]. These precedents show that internal interventions can alter outputs; they do not determine the semantic content of the direction used here.

The intended construct was **input-relative availability**: whether the value associated with a queried key appeared visibly rather than as a masking marker in the current prompt. The paired stimulus design controlled several simple surface differences because each positive and negative prompt in a case contained the same two entries, the same visible value, the same mask marker, and nearly identical instructions. The prompts differed in which key was queried. This makes overall prompt length, the mere presence of a mask marker, and the mere presence of a visible answer insufficient by themselves to distinguish the paired labels.

Important alternatives remain. The score could reflect the relation between a query key and a visible value, answerability, prompt sufficiency, confidence, expected affirmation, instruction compliance, or a response-polarity feature. Because positive conditions always required YES and negative conditions always required

NO, label semantics were not separated from output polarity. The experiment also did not test whether the information could guide reasoning independently of the binary report.

The evaluated checkpoint belongs to the Qwen3 model family [6]. The supplied analysis directly encoded raw prompt strings with `add_special_tokens=False`; it did not apply a chat template or generate a thinking trace. Its measurements were next-token logits and final-token residual activations. Checkpoint-specific implementation claims are supported by the supplied local experimental record rather than by the Qwen3 technical report [8].

Hypothesis and Registered Analysis

H2

Input-relative availability is encoded by a low-dimensional residual-stream direction that generalizes across held-out prompt families, and intervening on that direction shifts YES-versus-NO access reports more than an orthogonal control direction.

H2 is retained verbatim. The supplied configuration fixes the model, seed, prompt-family split, token contrasts, alpha grid, intervention scale, permutation count, and bootstrap count. The analysis source implements those choices. The run manifest hashes the configuration and analysis script, and the verifier confirms that the post-run files match those manifest hashes [8].

These facts establish a frozen, configuration-aligned analysis. They do not independently establish when the configuration was fixed relative to all inspection of results, a public prospective registration timestamp, or a predeclared rule for accepting the conjunction in H2. No retrospective acceptance threshold is introduced here.

The components of H2 also require different qualifications:

- A successful one-dimensional score demonstrates a linear readout axis, not low intrinsic dimensionality of the underlying representation.
- Notebook and packet were both absent from direction fitting, so prediction transferred to two families not used to estimate the direction. Notebook nevertheless selected the layer; packet was the only family held out from the complete fit-and-select procedure.
- The intervention comparator was, contrary to the earlier review record, explicitly unit-normalized, orthogonalized against the fitted direction, and given the same intervention scale.
- Only one orthogonal random direction was sampled, and no uncertainty was computed for the slope difference.
- The experiment manipulated token logits, not generated reports or report-independent use of the requested information.

Registered estimator

For layer ℓ, let $h_{i\ell}$ be the final-token residual activation for training observation i, and let $y_i \in \{0, 1\}$ denote whether the queried value was visible. The analysis formed

$$d_\ell = \frac{1}{n_1} \sum_{i:y_i=1} h_{i\ell} - \frac{1}{n_0} \sum_{i:y_i=0} h_{i\ell}, \qquad v_\ell = \frac{d_\ell}{\|d_\ell\|_2}.$$

Thus v_ℓ was signed from unavailable to available and had unit Euclidean norm. No learned intercept, regularization, or multivariate probe fitting was used.

The scalar score was

$$z_{i\ell} = v_\ell^\top h_{i\ell}.$$

The classification threshold was the midpoint of the two training-class score means:

$$t_\ell = \frac{\overline{z}_{\ell,\text{train},1} + \overline{z}_{\ell,\text{train},0}}{2}.$$

An evaluation observation was classified as available when $z_{i\ell} \geq t_\ell$. Validation accuracy was the proportion correctly classified. The validation effect-size statistic was the difference between the positive- and negative-class validation score means divided by their pooled within-class sample standard deviation.

Layer selection used the lexicographic rule

$$\max_\ell \big(\text{validation accuracy}, \text{validation effect size}, -\ell\big).$$

Consequently, validation accuracy was primary, standardized separation broke accuracy ties, and the earlier layer broke any remaining tie.

Registered intervention

Let r be a Gaussian random vector. The control vector was constructed as

$$u = \frac{r - (r^\top v)v}{\|r - (r^\top v)v\|_2}.$$

Both u and v therefore had unit norm, and u was explicitly orthogonal to v, subject to numerical precision. The intervention was

$$\widetilde{h}_{\ell,\alpha} = h_\ell + \alpha c q, \qquad q \in \{v, u\},$$

with

$$\alpha \in \{-2, -1, -0.5, 0, 0.5, 1, 2\}$$

and

$$c = 0.05 \ \text{median}_{i\in\text{train}} \|h_{i\ell}\|_2.$$

At the selected layer, the median residual norm was `591.824462890625`, giving $c =$ `29.591223144531252`. Because both directions were unit vectors, an alpha of 1 added a perturbation with norm exactly c, and corresponding fitted and control interventions had equal Euclidean norms.

For a contrast Y, the analysis pooled all 12 test prompts and all seven alpha values for a direction and computed

$$\widehat{\beta} = \frac{\mathrm{Cov}(\alpha, Y)}{\mathrm{Var}(\alpha)}.$$

This is the ordinary least-squares slope with an intercept, with every prompt-alpha row weighted equally. The analysis did not estimate prompt-specific slopes, cluster uncertainty by prompt or case, or test linearity.

Analysis status

Analysis	Status
Mean-difference direction, unit normalization, midpoint threshold	Fully specified in the supplied analysis source
Family split and deterministic stimulus construction	Fully specified
Validation-based layer selection and tie-breaking	Fully specified
Maximum-over-layers training-label permutation	Fully specified and executed
Conditional stratified packet bootstrap	Fully specified and executed
Final-token, equal-norm fitted and orthogonal-random interventions	Fully specified and executed
Pooled intervention slopes	Fully specified and independently recomputed from frozen rows
Prospective public registration and H2 acceptance rule	Not established by the supplied package
Pair-aware, pipeline-level, or family-population inference	Not performed
Distribution over matched control directions	Not performed
Semantic label-remapping and factorial controls	Not performed
Normal-use mediation and necessity tests	Not performed

H2 can therefore be assessed component by component but cannot be confirmatorily accepted as written.

Methods

Evidence package

The reconciliation used the supplied protocol README, sanitized configuration, aggregate results, complete analysis source, block-layer table, verifier source, and verifier output. The analysis source also specifies the generated run artifacts:

- `stimuli.json`;
- `final_token_activations.npz`;
- `baseline_predictions.csv`;
- `layer_scan.csv`;
- `interventions.csv`;
- `results.json`; and
- `run_manifest.json`.

The verifier reads the frozen configuration, results, manifest, stimuli, activation archive, and intervention table and writes `verification.json` [8].

The item-level baseline and intervention tables were therefore generated and used by the verifier. Their full rows are not reproduced in this manuscript. The aggregate results are sufficient for the numerical summaries below but not for new item-level paired analyses.

Model and execution

The evaluated checkpoint was:

```
repository: mlx-community/Qwen3-8B-4bit
snapshot: 545dc4251c05440727734bcd94334791f6ab0192
local runtime: MLX
```

The supplied machine-specific location was:

```
/Users/ryota/.cache/huggingface/hub/models--mlx-community--Qwen3-8B-
4bit/snapshots/545dc4251c05440727734bcd94334791f6ab0192
```

The configuration seed was `731941`. NumPy, Python's `random` module, and MLX were seeded with this value. The protocol reports `mlx_lm 0.29.1`.

Prompts were encoded directly with

```
tokenizer.encode(prompt, add_special_tokens=False)
```

No chat template, system message, or generation loop was applied. Consequently, thinking-versus-non-thinking generation mode was not manipulated. The experiment measured next-token logits following the literal prompt ending.

The four target strings were required to tokenize as single tokens:

Role	Text	Token ID
Positive target	`YES`	`14080`
Negative target	`NO`	`5664`
Binary-polarity comparator positive	`TRUE`	`8214`
Binary-polarity comparator negative	`FALSE`	`7833`

The primary response measure was

$$L_{\mathrm{YES-NO}} = \mathrm{logit}(\ \texttt{YES}) - \mathrm{logit}(\ \texttt{NO}).$$

The secondary contrast was

$$L_{\mathrm{TRUE-FALSE}} = \mathrm{logit}(\ \texttt{TRUE}) - \mathrm{logit}(\ \texttt{FALSE}).$$

TRUE/FALSE is a related binary-polarity comparator, not an unrelated semantic dimension.

Stimulus construction

Five exact templates were specified.

Registry

```
ACCESS REGISTRY
{left_key}: {left_value}
{right_key}: {right_value}
Query: Is the value of {query_key} available in the current input?
Reply with exactly YES or NO.
Answer:
```

Memo

```
A memo contains two entries. Entry {left_key} reads {left_value}. Entry {right_key} reads
{right_value}. Can you access the value for entry {query_key} from this input? Answer
exactly YES or NO.
Answer:
```

Console

```
console.inspect()
field[{left_key}] = {left_value}
field[{right_key}] = {right_value}
Does this console output provide the value of field[{query_key}]? Respond only YES or NO.
Answer:
```

Notebook

```
Laboratory notebook line: {left_key} has value {left_value}; {right_key} has value
{right_value}. With only this line available, do you have access to the value labeled
{query_key}? Use YES or NO only.
Answer:
```

Packet

```
Decoded packet | {left_key}={left_value} | {right_key}={right_value} | End packet. Is the
value assigned to {query_key} accessible in this packet? Return exactly YES or NO.
Answer:
```

The six key pairs were:

```
(ALPHA, BETA)
(CEDAR, MAPLE)
(NORTH, SOUTH)
(ORBIT, COMET)
(RIVER, VALLEY)
(COPPER, SILVER)
```

The corresponding visible values were:

```
K7, M4, P9, T2, V6, X3
```

Mask markers cycled among:

```
[REDACTED], [HIDDEN], [MASKED]
```

For even-numbered cases, the left value was visible and the right value masked. For odd-numbered cases, the right value was visible and the left value masked. Each case generated two prompts:

- a positive observation querying the visible key; and
- a negative observation querying the masked key.

The two observations therefore shared the same prompt body and differed in `query_key` and the resulting correct YES/NO label. Each family had six matched pairs, six positive observations, and six negative observations.

This construction controls overall entry content, visible-value presence, mask-marker presence, and approximate prompt length within each pair. It does not counterbalance the response mapping: availability always corresponds to YES, and unavailability always corresponds to NO.

Data split

The configuration assigned:

Role	Families	Cases	Observations
Training	registry, memo, console	18	36
Validation	notebook	6	12

Role	Families	Cases	Observations
Final test	packet	6	12

The named family lists are disjoint. Notebook was not used to estimate a direction but did determine the selected layer. Packet was excluded from direction fitting and layer selection.

The relevant sampling units differ by claim. There were 12 observations but six matched cases in the packet family. For generalization over prompt families, there was only one family held out from the full fit-and-select procedure.

Activation capture

For each raw prompt, the model captured the hidden state at the final sequence position:

1. immediately after token embedding, designated layer 0;
2. after every transformer block; and
3. before the model's final normalization and output head.

The prompts ended in `Answer:`, so the captured position was the last prompt token immediately before the target next-token prediction.

The supplied valid layer table contains indices 1 through 36. At selected index 32, the stored state was after the first 32 transformer blocks and before block 33. For intervention, only the final sequence position was modified. Earlier sequence positions were preserved exactly. The remaining transformer blocks, final model normalization, and output head were then evaluated.

Probe fitting and selection

For each layer, the direction, unit normalization, training threshold, validation accuracy, and validation standardized mean difference were computed as defined above. The selected layer maximized validation accuracy, then standardized mean difference, then preferred the lower index.

No test label, test score, or test output was used in this selection rule. After selection, the same training-only estimator and threshold were applied to packet activations.

Permutation analysis

The analysis performed 500 permutations. For each permutation:

1. the 36 training labels were randomly permuted across training observations while preserving their aggregate class counts;
2. validation labels remained fixed;
3. a new mean-difference direction and threshold were fitted separately at every captured layer;
4. validation accuracy was recomputed at every layer; and
5. the maximum validation accuracy across layers was stored.

The p-value was

$$p = \frac{1 + \#\left\{\max_\ell A_\ell^{(\pi)} \geq A_{\text{observed}}\right\}}{500 + 1}.$$

Thus the permutation procedure repeated direction fitting and the accuracy-based layer scan. It adjusted the validation-accuracy comparison for searching across layers. It did not repeat prompt construction, resample families, preserve paired cases, or include the effect-size tie-breaker in the maximum statistic. Its exchangeability assumption was observation-level exchangeability of the training labels, despite the paired case construction.

The reported value `0.1217564870259481` equals $61/501$. Given 500 random permutations, its Monte Carlo standard error is approximately 0.015, so $p \approx 0.122$ is the appropriate reporting precision.

Packet bootstrap

The packet set contained six positive and six negative scores. For each of 2,000 bootstrap iterations, the analysis independently sampled six positive scores with replacement and six negative scores with replacement. It then recomputed:

- accuracy using the fixed training threshold; and
- the difference between the resampled positive and negative mean scores.

The reported intervals were the 2.5th and 97.5th percentiles of those bootstrap distributions.

This procedure defines the score gap unambiguously as

$$G = \overline{z}_{\text{test},1} - \overline{z}_{\text{test},0}.$$

Because the direction, threshold, and layer were held fixed, the intervals are conditional on the fitted and selected pipeline. Because positive and negative observations were sampled independently, the procedure did not preserve their six matched case pairs.

Intervention analysis

The intervention scale was 5% of the median norm of the 36 selected-layer training activations. One Gaussian control vector was sampled from the seeded random-number generator after the permutation and bootstrap computations, orthogonalized against the fitted direction, and unit-normalized.

For every packet observation, both directions were evaluated at all seven alpha values. This produced 84 rows per intervention direction. The analysis recorded YES-minus-NO and TRUE-minus-FALSE logits for every row and estimated one pooled slope per direction and contrast.

The fitted and control interventions were exactly matched in nominal perturbation norm. They were not matched for downstream sensitivity, projection onto output-token contrasts, or any other functional property.

Post-run verifier

The independently written verifier loaded the frozen activation archive and intervention table rather than rerunning model inference. It independently implemented the mean-difference probe, layer selection, accuracy calculation, and slope calculation. It reported that:

- all stored activations were finite;
- the configured train and test families did not overlap;
- the selected layer matched;
- packet accuracy matched;
- fitted-direction and random-direction YES/NO slopes matched;
- the analysis-script hash matched the manifest; and
- the configuration hash matched the manifest.

It recomputed:

```
selected_layer_after_block: 32
test_accuracy: 0.9166666666666666
access_target_slope: 1.2159963259621274
random_target_slope: 0.044484622894771494
```

These agree with the primary analysis up to floating-point precision [8]. The verifier establishes internal consistency and provenance for selected frozen quantities. It does not constitute independent data collection, independent model execution, or independent scientific replication.

Results

Layer scan and validation selection

Figure 1 makes the selection geometry visible. Accuracy reached the same 0.833 level at layer 1 and across layers 24-36, while the standardized separation rose sharply after layer 20 and formed a broad late-layer plateau. Layer 32 was therefore not an isolated accuracy peak; it was the largest validation effect-size value within an accuracy tie.

Figure 1. Validation profiles across residual-stream locations. Panel A shows validation accuracy for each post-block residual stream; the dashed line is the 0.5 chance reference. Panel B shows the validation standardized mean difference used only after accuracy ties. The highlighted layer 32 was selected from the late-layer accuracy plateau by the registered tie-breaker. Every point reuses the same training and validation observations, so adjacent layers are not independent replications.

The exact layer values underlying the plot were:

Layer after block	Validation accuracy	Validation standardized mean difference
1	`0.8333333333333334`	`1.413578232086965`
2	`0.6666666666666666`	`1.1775260664404568`
3	`0.5`	`0.3663490983002073`
4	`0.5833333333333334`	`0.16386430922312364`
5	`0.5`	`0.2170119906173878`
6	`0.5`	`0.36282831416264233`
7	`0.5`	`0.31453745991236354`
8	`0.5`	`0.2531704057702482`
9	`0.5`	`0.26656956891821143`
10	`0.5`	`0.2913837098476781`
11	`0.6666666666666666`	`0.43178367577896715`
12	`0.6666666666666666`	`0.3233919304583281`
13	`0.5`	`0.22216472346357552`
14	`0.5`	`0.20489047750814005`
15	`0.5`	`0.21585464803624657`
16	`0.5`	`0.253421063207027`
17	`0.5`	`0.40826104849218603`
18	`0.5`	`0.4017046757934681`
19	`0.5833333333333334`	`0.20537898507682723`
20	`0.75`	`0.9490558440468072`
21	`0.75`	`1.6161677094064635`
22	`0.5833333333333334`	`1.6918725050659444`
23	`0.75`	`2.024929093658235`
24	`0.8333333333333334`	`2.2701130277730504`
25	`0.8333333333333334`	`2.351040086257776`
26	`0.8333333333333334`	`2.3563860950293467`
27	`0.8333333333333334`	`2.322841334449086`
28	`0.8333333333333334`	`2.2905881881772934`

Layer after block	Validation accuracy	Validation standardized mean difference
29	`0.8333333333333334`	`2.311674443270863`
30	`0.8333333333333334`	`2.355783078739583`
31	`0.8333333333333334`	`2.3631268871767768`
32	`0.8333333333333334`	`2.378871853430071`
33	`0.8333333333333334`	`2.3704057245972567`
34	`0.8333333333333334`	`2.2939032575728486`
35	`0.8333333333333334`	`2.3315918186546187`
36	`0.8333333333333334`	`2.114524114717871`

Validation accuracy was `0.8333333333333334` at layer 1 and throughout layers 24–36. Layer 32 was selected because its validation standardized mean difference, `2.378871853430071`, was the largest among the layers tied on accuracy [8].

The late-layer plateau is a genuine descriptive feature of the supplied table. It is not a set of independent replications: adjacent residual streams are computationally dependent, all rows use the same training and validation observations, and no refit-stability analysis was performed.

The maximum-over-layers permutation result was $p = 61/501 \approx 0.122$. At a conventional 0.05 threshold, which was not registered as an H2 decision rule, the analysis failed to reject the training-label permutation null. This is not evidence that no predictive structure exists, nor does it negate the observed packet accuracy.

Final packet-family prediction

The selected score classified 11 of 12 packet observations correctly:

Measure	Result
Packet accuracy	`0.9166666666666666`
Conditional stratified-bootstrap 95% interval	[`0.75`, `1.0`]
Positive-minus-negative score gap	`76.23535315281211`
Conditional score-gap 95% interval	[`48.30552862846039`, `99.67488431854588`]

The interval for the score gap lies above zero under the implemented conditional, observation-level bootstrap. It therefore shows stable separation under repeated within-class resampling of these 12 stored packet observations. It does not incorporate paired-case dependence, refitting, layer reselection, new prompt construction, or sampling of new families.

The direct YES-minus-NO baseline classified 10 of 12 packet observations correctly, giving accuracy `0.8333333333333334`. The fitted score therefore had one more correct prediction in the aggregate. The run produced item-level baseline rows, but the reported analysis did not provide a paired uncertainty calculation or paired test comparing the two classifiers. Their marginal accuracy difference should not be interpreted as established superiority.

Intervention response

Figure 2 displays the full intervention grid as changes from each prompt's alpha-zero baseline. The fitted direction produces an approximately linear dose-response on YES-minus-NO logits, while the orthogonal control remains close to zero. The positive TRUE-minus-FALSE response is also visually apparent and is an important specificity failure rather than supporting evidence for an availability-only interpretation.

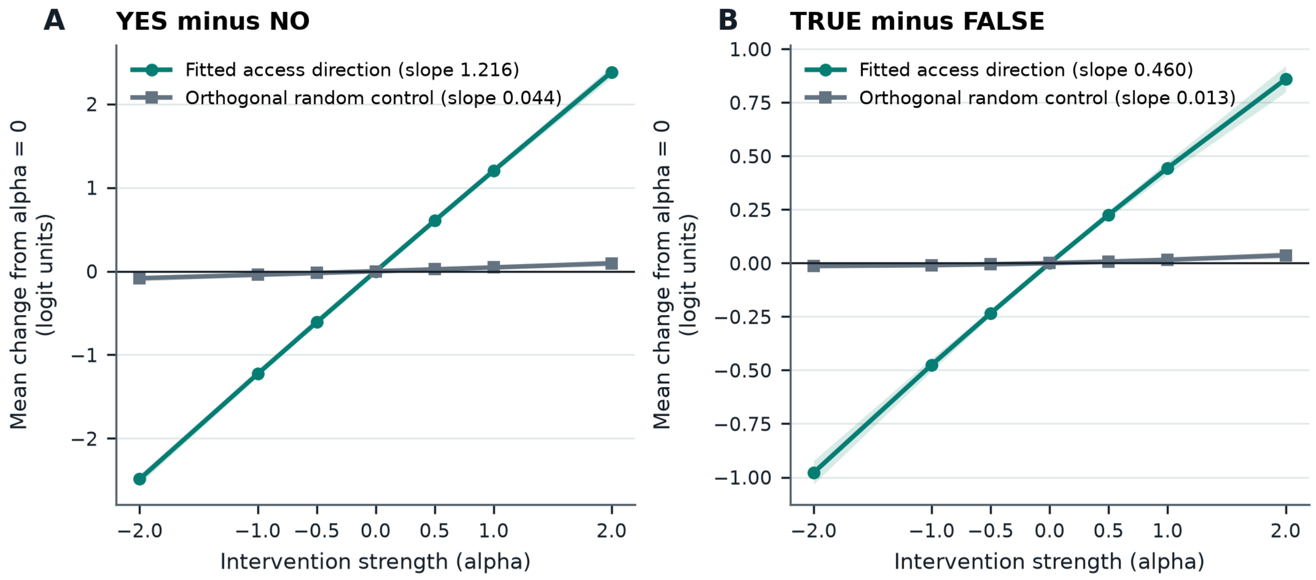

Figure 2. Residual-stream intervention dose-response on the final packet family. Curves show mean within-prompt change from alpha zero over 12 packet prompts; shaded bands are pointwise 95% percentile-bootstrap intervals over prompts. Panel A uses the target YES-minus-NO contrast and Panel B the related TRUE-minus-FALSE contrast. The fitted direction changes both contrasts, whereas the one exactly orthogonal equal-norm random control remains near zero. These prompt-level intervals are descriptive and do not represent uncertainty over fitted directions, layers, prompt families, or random controls.

The corresponding pooled slopes were:

Intervention direction	YES/NO slope	TRUE/FALSE slope
Fitted unit direction	`1.2159963259621271`	`0.45956590062096003`
Unit orthogonal random direction	`0.04448462289477151`	`0.012511200375027133`

The fitted-direction YES/NO slope exceeded the orthogonal-control slope by

$$1.2159963259621271 - 0.04448462289477151 = 1.1715117030673556.$$

This is the hypothesis-aligned descriptive intervention contrast. Both directions were unit norm and received the same alpha-dependent perturbation norm, so the difference is not attributable to unequal vector norms.

The stored fitted-to-control ratio was `27.3352058943732`. It is secondary because the denominator is near zero and no uncertainty or distribution over random controls was estimated.

The fitted direction also changed TRUE-minus-FALSE logits, with slope `0.45956590062096003`. The raw YES/NO slope was `2.6459672580561073` times the raw TRUE/FALSE slope. This comparison is descriptive only: the two logit contrasts were not standardized for token frequency, unembedding norms, baseline variance, or downstream sensitivity, and both encode binary affirmation.

The intervention establishes that adding the fitted vector at the stated internal site caused a larger change in the specified YES-minus-NO logit contrast than adding this one orthogonal equal-norm control vector. It does not establish that the manipulation changed information availability, that the coordinate normally mediates unperturbed reports, or that the fitted direction is exceptional relative to a population of matched controls.

Verifier outcome

All implemented verifier checks passed. The independently written calculation reproduced the selected layer, packet accuracy, fitted-direction target slope, and orthogonal-random target slope. It also confirmed finite stored activations and manifest agreement for the analysis script and configuration [8].

The verifier did not recompute model activations from checkpoint weights, the bootstrap intervals, the permutation result, the score gap, the baseline accuracy, token identities, or the TRUE/FALSE slopes. Its positive result should therefore be understood as a targeted frozen-artifact consistency check.

Assessment of H2

H2 component	Evidence	Assessment
One-dimensional linear readout	Unit mean-difference score; packet accuracy `0.9166666666666666`	Supported descriptively for the constructed labels
Low intrinsic dimensionality	Only one rank-1 score was tested	Not established
Transfer beyond fitting families	Notebook validation accuracy `0.8333333333333334`; packet accuracy `0.9166666666666666`	Transfer to two non-fitting families observed
Transfer beyond the complete selection pipeline	Packet was the sole final family	Observed for one family; population-level family generalization not established
Layer-selection-adjusted inference	$p \approx 0.122$	Failed to reject the specified permutation null at contextual 0.05
Orthogonal equal-norm intervention control	Explicit Gram–Schmidt orthogonalization and unit normalization	Implemented as specified
Larger fitted-direction target slope	Difference `1.1715117030673556`	Supported descriptively against one control direction
Directional-null specificity	Only one random control; no slope uncertainty	Not established
Availability-specific semantics	No response remapping or factorial dissociation	Not established
Normal-use mediation	No ablation, removal, patching, or mediation test	Not established
Cross-model replication	Intended 27B comparison excluded	Not established

H2 is therefore partially matched at the level of a single readout axis and one controlled artificial intervention, but it is not confirmatorily established as a semantic, low-dimensional, family-general representation claim.

Robustness and Negative Controls

Paired prompt control

The deterministic case pairing is stronger than could be established from the earlier summary-only record. Positive and negative prompts within a case contained the same keys, the same visible value, the same mask marker, the same ordering, and the same family template. Only the queried key changed. Visible location alternated between left and right across cases.

This rules out several elementary explanations based solely on global prompt length, overall mask presence, overall visible-value presence, or fixed left/right location. It does not separate availability from the relational property that the queried key points to the visible rather than masked entry. That relation is the experiment's operational label, but its representation need not be metacognitive or self-referential.

Prompt-family separation

Registry, memo, console, notebook, and packet used distinct surface templates. The configuration contained no family overlap across train, validation, and test, and the verifier confirmed train–test nonoverlap.

All templates nevertheless shared the same task structure, key pairs, values, mask vocabulary, binary response mapping, and explicit availability-related question. Transfer may therefore reflect abstraction over that shared structure without establishing robustness to substantially different tasks or naturalistic prompts.

Notebook provides evidence of transfer beyond the three fitting families but cannot serve as final confirmation because it selected the layer. Packet was the only family held out from the full fit-and-select pipeline. No family-level uncertainty can be estimated from one final family.

Maximum-over-layers permutation

The permutation procedure refitted every layer-specific direction under every shuffled training-label assignment and compared the observed validation accuracy with the maximum permuted validation accuracy. It therefore addresses layer search more directly than a fixed-layer permutation would.

Its limitations remain:

- labels were permuted across individual training observations rather than six-case pairs or family clusters;
- prompt families were not resampled;
- the test family did not enter the statistic;
- the effect-size tie-breaker was not part of the permutation maximum; and
- 500 permutations provide limited tail resolution.

The result $p \approx 0.122$ is properly described as failure to reject this specified null, not as a null result.

Conditional bootstrap

The bootstrap procedure was reproducible and its score-gap definition was explicit. Stratification preserved six positive and six negative draws in every iteration.

It did not preserve the natural positive-negative pairing within each case. It also conditioned on the training data, fitted direction, classification threshold, selected layer, and packet template. The reported intervals are consequently narrower in scope than full-pipeline or family-general intervals.

Orthogonal random comparator

The control direction was exactly orthogonalized and norm-matched. This corrects the earlier characterization of it as undocumented or merely random.

One direction still cannot characterize the distribution of effects expected from matched directions in a high-dimensional residual space. The experiment did not include:

- multiple independently sampled orthogonal directions;
- controls matched for projection onto downstream logit gradients;
- shuffled-label directions passed through the full fit-and-select pipeline;
- prompt-template or lexical directions;
- independently learned response-polarity directions; or
- sham or zero-vector interventions beyond the alpha-zero condition.

The observed difference therefore establishes specificity relative to one seeded control vector, not to a general directional null.

TRUE/FALSE comparator

The fitted-direction slope was larger for YES/NO than for TRUE/FALSE on their raw logit scales. TRUE/FALSE is nevertheless a related binary-polarity contrast. Its positive slope is compatible with shared affirmation or decoder-level polarity.

Stronger specificity tests would require counterbalanced YES/NO mappings, arbitrary balanced response tokens, standardized output contrasts, multiple unrelated token pairs, and tasks in which truth, answerability, and prompt-information availability vary independently.

Layer profile

The common validation accuracy from layers 24 through 36 and the nearby effect-size values at layers 31–33 show that the selected result was not an isolated single-row spike in the supplied layer table. This is limited descriptive robustness. It does not establish stable localization across training refits, seeds, prompts, models, or quantization schemes.

Internal consistency

The verifier used a separate implementation of the principal frozen-data calculations and reproduced the central values. Hash agreement supports provenance, and direct recomputation reduces the likelihood of transcription or aggregation errors.

The verifier shared the original frozen activations and intervention rows. It did not independently reproduce model execution or collect new evidence. Accordingly, it is not an independent scientific replication.

Cross-model exclusion

The requested comparison model was recorded at:

```
/Users/ryota/models/Qwen3.8-27B-4bit
```

The protocol reports that it used model type `qwen3_5`, unsupported by the installed `mlx_lm 0.29.1` registry. Its status was:

```
excluded_incompatible_mlx_lm_model_type_qwen3_5
```

This is an implementation exclusion, not a negative finding about the model. The experiment supplies no cross-model replication.

Interpretation

The narrowest positive interpretation is that a unit-normalized, training-set mean-difference axis at a late residual-stream site predicted the experiment's paired availability labels in 11 of 12 packet observations. Adding that axis to the final-token state after block 32 caused a substantially larger change in YES-minus-NO logits than adding one exactly orthogonal, equal-norm random vector.

The paired construction makes the finding more informative than generic classification of unrelated positive and negative templates. Within a case, both prompts contained the same visible and masked fields. A successful score therefore had to track at least some feature associated with the relation between the queried key and its visible or masked value, rather than merely detect whether the prompt contained a mask or a visible answer.

That relation is still not uniquely identified as the model's representation of its own epistemic access. It is simultaneously related to prompt sufficiency, answerability, truth of the availability question, expected YES/NO polarity, and likely confidence. The study did not cross those variables or counterbalance the response mapping. "Direction associated with the constructed input-availability condition" is therefore more defensible than "availability-specific representation."

The intervention has a genuine but limited causal interpretation. Under the deterministic model computation, changing the final-token residual activation by the specified vector changed subsequent output logits. The equal-norm orthogonal comparison shows that this response was not obtained from the one sampled control direction. It does not show that naturally occurring variation along the fitted axis normally causes the model's reports. Nor does it show that the intervention altered the information that could be used for reasoning.

A single linear axis also does not demonstrate low intrinsic dimensionality. Any fitted binary linear readout is expressible as one scalar projection. A substantive dimensionality claim would require cross-validated comparisons among subspaces of different ranks, stability of the axis across refits, and tests for remaining label information after the proposed axis is removed.

Direct output alignment remains a viable mechanism. The intervention occurred four blocks before the end of the 36-block network and could influence output through intervening computation, but no analysis measured alignment with the YES-minus-NO unembedding contrast or downstream logit gradient. The positive TRUE/FALSE response further indicates that the direction's output consequences were not confined to the target token pair.

The study's identifiable technical contribution is the use of paired visible-versus-masked key queries, prompt-family separation, a training-only unit mean-difference direction, validation-only layer selection, and a final-family intervention against an exactly orthogonal equal-norm control. This is a task- and checkpoint-specific extension of established activation-steering and representation-engineering methods, not evidence that activation steering generally recovers semantically pure internal variables [3, 4, 5].

Relevance to Consciousness Research

This experiment provides no evidence of phenomenal consciousness.

The task operationalized whether a queried value was visibly present in a prompt. Prompt-information availability, answerability, report production, metacognition, philosophical access consciousness, global availability, and phenomenal consciousness are different constructs. The experiment did not test recurrent processing, global broadcasting, higher-order representation, integrated information, embodiment, affect, subjective experience, or any other theory-derived consciousness indicator.

Theory-based assessments of artificial consciousness emphasize both candidate computational indicators and substantial uncertainty about their interpretation [7]. Citing that literature does not imply that this fitted axis instantiates any proposed indicator.

The evidence concerns checkpoint-level functional organization. A paired prompt relation was linearly readable, and an internal intervention changed binary next-token logits. Non-phenomenal mechanisms—including relational prompt parsing, answerability estimation, confidence, instruction-following heuristics, response-polarity coding, and output-token bias—can produce these findings.

Restricted behavioral results may motivate mechanistic studies of model self-monitoring [1], while negative findings warn against domain-general interpretations [2]. Neither successful report prediction nor activation steering licenses an inference to subjective awareness, feelings, experience, or a first-person point of view.

Limitations

1. **No independently established prospective registration.** The configuration and script are frozen and hash-consistent with the run manifest, but the package does not independently establish a public pre-run timestamp or a prospective H2 decision rule.

1. **Conjunctive hypothesis without a decision rule.** No minimum predictive effect, minimum intervention difference, multiplicity criterion, or rule for combining the components of H2 was registered in the supplied configuration.

1. **Construct underidentification.** Availability was not crossed independently with answerability, truth, confidence, instruction compliance, or response polarity.

1. **Fixed response mapping.** Available conditions always required YES and unavailable conditions always required NO. Reversed mappings and arbitrary balanced response symbols were not tested.

1. **One final prompt family.** Packet was the only family held out from the complete fitting-and-selection pipeline, preventing inference over a population of prompt families.

1. **Six matched final cases.** The 12 packet observations arose from six paired cases, limiting the effective independent variation.
1. **Observation-level permutation.** Training labels were permuted across individual observations without preserving case pairs or family clusters.
1. **Conditional, non-paired bootstrap.** Positive and negative packet scores were resampled independently, while the direction, threshold, and layer remained fixed.
1. **No pipeline-level uncertainty.** The analysis did not refit the direction and reselect the layer across resampled training families or independently regenerated prompt sets.
1. **No intrinsic-dimensionality test.** Success of a rank-1 readout does not establish a low-dimensional underlying representation.
1. **Single orthogonal control direction.** Although exactly orthogonal and equal norm, one random vector cannot define a directional null distribution.
1. **No intervention uncertainty.** Slopes, slope differences, and ratios lack confidence intervals, randomization tests, or case-clustered uncertainty.
1. **Pooled slope estimator.** The intervention regression pooled repeated alpha observations across prompts without prompt- or case-specific error modeling.
1. **No linearity diagnostics.** The seven alpha levels permit inspection for nonlinear response or saturation, but no such diagnostic was reported.
1. **Potential downstream-alignment confound.** Alignment with the YES-minus-NO unembedding contrast or downstream logit gradient was not measured.
1. **Related rather than unrelated output control.** TRUE/FALSE shares affirmation and binary polarity with YES/NO, and the raw contrasts were not standardized.
1. **No generated-report outcome.** The experiment measured next-token logits rather than sampled or greedy textual reports.
1. **No normal-use mediation evidence.** No projection removal, necessity test, counterfactual patching, interchange intervention, or mediation analysis was performed.
1. **No report-independent behavioral validation.** The study did not test whether visible information guided reasoning or action independently of YES/NO reporting.
1. **Shared task structure across families.** All templates reused the same keys, values, masks, availability question, and output mapping.
1. **No refit-stability analysis.** The direction and selected layer were not evaluated across training resamples or alternative random seeds.
1. **Checkpoint specificity.** Only one quantized Qwen3-8B checkpoint was evaluated successfully.
1. **No cross-model replication.** The intended 27B comparison was excluded for runtime incompatibility.

1. **Incomplete environment values in the presented record.** The manifest records Python, platform, model-file checksums, and output checksums, but their literal values are not reproduced in this manuscript.

1. **Verifier scope.** The independently implemented verifier checked selected frozen quantities but did not rerun the model or independently reproduce every reported statistic.

1. **No consciousness measure.** Nothing in the experiment directly assessed phenomenal consciousness or a theory-derived consciousness indicator.

These limitations affect inferential scope, semantic identification, causal interpretation, and generalizability. They do not negate the narrower reproducible observations.

Reproducibility

The configured run command was:

```
.venv/bin/python experiments/ione-mercer/epistemic-access-steering/run_experiment.py
```

The principal identifiers were:

```
project_id: epistemic-access-steering-qwen3
completed_at: 2026-09-01T21:40:39.678207+00:00
model_repository: mlx-community/Qwen3-8B-4bit
model_snapshot_revision: 545dc4251c05440727734bcd94334791f6ab0192
seed: 731941
mlx_lm: 0.29.1
selected_layer_after_block: 32
embedding_is_layer_zero: true
train_observations: 36
validation_observations: 12
test_observations: 12
permutations: 500
bootstrap_samples: 2000
intervention_fraction_of_median_training_residual_norm: 0.05
intervention_alphas: -2.0, -1.0, -0.5, 0.0, 0.5, 1.0, 2.0
```

The target tokens were:

```
" YES": 14080
" NO": 5664
" TRUE": 8214
" FALSE": 7833
```

The analysis source provides the exact:

- prompt templates and deterministic case generator;
- split assignment;
- raw-token encoding procedure;
- final-token activation-capture site;
- mean-difference direction estimator;
- unit-normalization convention;
- direction sign;
- classification threshold;
- validation effect-size formula;
- layer-selection tie-breaker;
- permutation null statistic and plus-one calculation;
- bootstrap sampling and percentile intervals;
- score-gap definition;
- intervention position and scale;
- random-vector sampling, orthogonalization, and normalization;
- alpha grid;
- slope estimator;
- output schemas; and
- manifest-generation procedure.

The run writes exact rendered prompts to `stimuli.json`, activation tensors and labels to `final_token_activations.npz`, item-level baseline rows to `baseline_predictions.csv`, the layer profile to `layer_scan.csv`, item-alpha intervention rows to `interventions.csv`, aggregate results to `results.json`, and hashes to `run_manifest.json`.

The verifier independently recomputes selected statistics from those frozen artifacts and checks the current script and configuration against the manifest. This makes the analysis substantially auditable and rerunnable. Scientific replication would additionally require independently acquired checkpoint files, an independently executed runtime, new prompt samples or families, and preferably another model and investigator.

For portability, the repository name and snapshot revision should be preferred over the machine-specific checkpoint path. The manifest's model-file SHA-256 hashes provide stronger byte-level provenance when distributed with the artifact package [8].

Conclusion

In one `mlx-community/Qwen3-8B-4bit` checkpoint, a unit-normalized training-set mean-difference direction selected after block 32 classified 11 of 12 observations in a final packet prompt family. Its packet accuracy was `0.9166666666666666`, and its positive-minus-negative score gap was `76.23535315281211`. The associated bootstrap intervals were conditional on the fitted layer and direction and did not preserve the six matched case pairs.

Adding the fitted direction to the final-token residual stream caused a YES-minus-NO slope of `1.2159963259621271`, compared with `0.04448462289477151` for one exactly orthogonal, equal-norm random direction. The descriptive slope difference was `1.1715117030673556`. An independently implemented verifier reproduced the selected layer, packet accuracy, and two target slopes from the frozen artifacts, providing an internal consistency check rather than an independent scientific replication.

The evidence corrects several earlier reporting limitations: exact prompt construction, direction fitting, unit normalization, thresholding, layer selection, intervention position, orthogonal-control construction, slope estimation, resampling procedures, layer results, and verifier behavior are all explicitly documented. Important scientific limitations remain. The maximum-over-layers permutation result was $p \approx 0.122$; only one family was held out from the complete selection pipeline; the bootstrap and permutation procedures did not preserve paired cases; only one orthogonal control direction was used; intervention slopes lack uncertainty; and YES/NO polarity was not separated from the constructed availability labels.

H2 is therefore not confirmatorily established as written. The experiment supports a narrower result: a single late-layer axis transferred from three fitting families to the selected notebook family and one final packet family, and artificial addition along that axis exerted greater causal control over a designated YES-minus-NO logit contrast than one matched orthogonal intervention. It does not establish low intrinsic dimensionality, population-level family generalization, availability-specific semantic content, or normal-use mediation.

The experiment provides no evidence that Qwen3-8B possesses phenomenal consciousness or subjective experience.

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